AI Analytics Assistant: Natural Language to Safe SQL

Project Proposal | AWS Machine Learning Course | Prepared by: [Mohamed El-Morshdy - Mohamed Hany – Belal Wael – Abdallah Salama]

1. Project Proposal

1.1 Overview

Managers and decision-makers often depend on data analysts to answer even simple questions about their data, because they do not know SQL. This slows down decisions, and poorly written queries can return wrong results or even damage data.

The AI Analytics Assistant lets any user ask a question in plain English, such as "What are the total sales for each month in 2026?", and receive a verified answer, a chart, and a short explanation. The system uses a Large Language Model (Amazon Bedrock) to translate the question into SQL, validates the SQL for safety, runs it on an AWS data lake, and presents the result visually.

Pipeline: Question → LLM → SQL Safety Check → Database → Result → Visualization → Explanation

1.2 Objectives

- Allow non-technical users to query data using natural language.
- Guarantee that only safe, read-only queries are executed (no DELETE, DROP, UPDATE, etc.).
- Enforce access control so each user only sees the data they are permitted to see.
- Automatically choose and display a suitable chart for each result.
- Explain to the user how the answer was produced, to build trust.
- Apply AWS cloud, data engineering, and machine learning concepts from the course in one working solution.

1.3 Scope

In scope

- Natural language to SQL generation using Amazon Bedrock.
- SQL safety validation layer (read-only, allow-listed tables and columns, row limits).
- Data lake on Amazon S3 with AWS Glue and Amazon Athena.
- Role-based access control for users.
- Automatic visualization and plain-language query explanation.
- A web interface for asking questions and viewing results.

Out of scope

- Writing or modifying data through the assistant.
- Training a custom LLM from scratch.
- Real-time streaming data ingestion.
- Support for languages other than English (possible future work).

1.4 Proposed Architecture

Layer	AWS Service	Purpose
Storage	Amazon S3	Data lake for raw and curated datasets
Data catalog and ETL	AWS Glue	Schema discovery, cleaning, and transformation
Query engine	Amazon Athena	Runs SQL directly on S3 data
AI	Amazon Bedrock	Converts questions to SQL and writes explanations
Application logic	AWS Lambda and Amazon API Gateway	Orchestration and SQL safety validation
Security	AWS IAM, Amazon Cognito,	Authentication and per-user

Layer	AWS Service	Purpose
	Lake Formation	data access
Interface	Streamlit or web app (hosted on AWS)	Question input, charts, explanations
Monitoring	Amazon CloudWatch	Logs, metrics, and alerts

2. Project Plan

2.1 Timeline (8 Weeks)

Phase	W1	W2	W3	W4	W5	W6	W7	W8
Requirements and design	■							
Data lake setup (S3, Glue, Athena)		■	■					
LLM integration (Bedrock)			■	■				
SQL safety and access control				■	■			
Visualization and explanation					■	■		
Testing and evaluation						■	■	

Documentation and final demo							■	■

2.2 Milestones and Deliverables

Milestone	Week	Deliverable
M1: Design approved	1	Architecture diagram, requirements document
M2: Data lake ready	3	Sample dataset queryable through Athena
M3: NL-to-SQL working	4	Bedrock generates correct SQL for sample questions
M4: Security complete	5	SQL validator and role-based access in place
M5: Full prototype	6	End-to-end flow with charts and explanations
M6: Evaluation complete	7	Test report with accuracy and performance results
M7: Final delivery	8	Live demo, final report, and presentation

2.3 Resource Allocation

- **People:** 4 team members (see Section 3).
- **Cloud:** AWS account with S3, Glue, Athena, Bedrock, Lambda, API Gateway, Cognito, and CloudWatch.
- **Data:** A public or synthetic sales dataset (for example, orders, customers, products).
- **Cost control:** AWS Budgets alert, Athena workgroup query limits, and small test datasets.

3. Task Assignment and Roles

Role	Member	Responsibilities
Project Manager and Integration Lead	[Name]	Planning, coordination, final integration, documentation
Data Engineer	[Name]	S3 data lake, Glue ETL and catalog, Athena tables
AI / Prompt Engineer	[Name]	Bedrock integration, prompt design, SQL accuracy, explanations
Backend and Security Engineer	[Name]	SQL safety validator, IAM, Cognito, Lake Formation, Lambda and API
Frontend and Visualization Developer	[Name]	User interface, automatic chart selection, result display

Testing and documentation are shared by the whole team. Roles can be combined if the team is smaller.

4. Risk Assessment and Mitigation Plan

Risk	Likelihood	Impact	Mitigation
LLM generates incorrect SQL	Medium	High	Provide schema context in prompts, use example queries, show the SQL and explanation to the user, test on a question set
Dangerous or	Medium	High	Allow only SELECT

Risk	Likelihood	Impact	Mitigation
malicious queries (DROP, DELETE, injection)			statements, parse and validate SQL, use a read-only IAM role
Users access data they should not see	Low	High	Role-based access with Lake Formation and IAM, filter allowed tables and columns
High AWS costs	Medium	Medium	Budget alerts, Athena scan limits, partitioned and compressed data (Parquet)
Slow response time	Medium	Medium	Optimize data format, limit rows, cache frequent queries
Ambiguous user questions	High	Medium	Ask a clarifying question or show the assumptions used
Team delays or limited time	Medium	Medium	Weekly check-ins, prioritize core features first, keep extra features optional

5. Key Performance Indicators (KPIs)

KPI	Target
SQL accuracy (correct results on a test set of 30+ questions)	85% or higher
Response time (question to result)	Under 10 seconds
Dangerous queries blocked	100%
Access control correctness (unauthorized data never returned)	100%
System uptime during demo and testing	99% or higher
Chart suitability (appropriate chart chosen automatically)	85% or higher
User adoption and satisfaction (pilot users who find it useful)	80% or higher

6. Expected Outcome

A working cloud-based assistant that turns plain-English questions into safe, verified SQL, returns results with a suitable chart and a clear explanation, and demonstrates the practical use of AWS cloud services, data engineering, and generative AI in a single project.